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Deep learning model for real time moisture content detection and prediction in white tea withering using near infrared spectroscopy

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This study proposes a novel deep learning model, named STA-BiGRU-XGBoost, for predicting moisture content during the white tea withering process using near-infrared spectroscopy. The model integrates spatiotemporal attention mechanisms, bidirectional gated recurrent units (BiGRU), and the XGBoost algorithm to address challenges such as extended withering durations, environmental variability, and temporal variations in spectral data. The maximum relevance minimum redundancy (mRMR) algorithm is employed to select critical variables, including hot air velocity, air duct temperature, and spectral absorbance. A spatial attention mechanism enhances feature relevance, while BiGRU captures long-term temporal dependencies. Temporal attention further adjusts the weights of key time steps. XGBoost was incorporated to improve prediction stability under the investigated production-line conditions. Experimental results obtained from a production-line dataset collected from Zhenghe County during the March–May 2025 production season show that the STA-BiGRU-XGBoost model achieved the best performance among the compared models, with RMSE = 0.0920, MAE = 0.0772, and R^2 = 0.9806. Furthermore, the model's interpretability is validated through attention weight visualization, highlighting key features associated with moisture evaporation dynamics. It should be noted that the current validation was conducted within a specific production season and fresh-leaf source region; broader generalization across different geographic origins, cultivars, and extreme weather conditions requires further cross-origin and multi-season validation.

Keywords White tea withering, Real-time moisture content detection and prediction, Near-infrared spectroscopy, Deep learning prediction model, Spatiotemporal attention mechanism

White tea, one of China's six major tea types, is renowned for its minimal processing and natural flavor. Its quality is largely determined during the withering process, in which tea leaves undergo moisture loss alongside complex biochemical transformations. These changes significantly influence flavor, aroma, and overall quality, making moisture content a crucial indicator of withering degree and product stability. Moreover, moisture content affects the conversion of polyphenolic compounds and the retention of amino acids, which contribute to the characteristic flavor and aroma of white tea[1–3].

The withering process typically lasts 48–72 h and is influenced by factors such as raw material variety, harvest season, leaf maturity, and initial moisture content. Fluctuations in moisture levels can lead to substantial variations in water loss rates and enzymatic activity, ultimately affecting product consistency. Currently, the assessment of withering degree relies primarily on subjective observations, such as leaf softness, color changes, and aroma, which are insufficient for large-scale standardized production[4]. Therefore, a real-time, non-destructive, and accurate method for monitoring moisture content during withering is needed.

Traditional moisture content measurement methods, such as oven drying and Karl Fischer titration, offer high accuracy but are time-consuming, destructive, and unsuitable for online monitoring. Recent advancements in non-destructive testing, particularly near-infrared spectroscopy (NIRS), have demonstrated considerable

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potential for real-time moisture detection in agricultural products due to their sensitivity to molecular vibrations, including those of water molecules[5]. NIRS has been successfully applied in processes such as green tea fixation, black tea withering, and tea drying, although challenges remain for prolonged processes like white tea withering[6,7].

However, unlike relatively short and concentrated processing stages such as green tea fixation or drying, white tea withering is characterized by a long duration, significant dynamic changes in environmental conditions, and strong process non-stationarity. Continuous variations in environmental temperature, humidity, ventilation conditions, and leaf pile status during withering lead to considerable temporal variations in both moisture migration within tea leaves and their near-infrared spectral responses. Recent studies indicate that during extended processing periods, moisture migration and environmental fluctuations can cause nonlinear drift and dynamic instability in near-infrared spectra, significantly undermining the predictive stability of models based on static assumptions[8,9]. Particularly in long-duration monitoring scenarios such as tea withering, spectra collected at different times may exhibit substantial differences even at similar moisture levels, imposing higher demands on model generalization and robustness[10].

To address these challenges, existing studies primarily employ traditional chemometric methods such as Partial Least Squares Regression (PLSR) and Support Vector Regression (SVR), or spectral correction techniques like External Parameter Orthogonalization (EPO) and Generalized Least Squares Weighting (GLSW) to mitigate environmental disturbances[11,12]. However, these methods generally rely on stable operational assumptions or additional environmental parameter inputs, making them difficult to implement consistently over the long term in actual white tea withering production lines. Furthermore, traditional modeling approaches struggle to simultaneously capture the nonlinear characteristics of moisture information and its temporal evolution in high-dimensional spectral data, limiting their applicability to prolonged withering processes.

In recent years, deep learning methods have gained significant attention in spectral analysis and food processing monitoring. Convolutional Neural Networks (CNNs) can automatically extract multi-scale spectral features, while Recurrent Neural Networks (RNNs) and their variants, such as Long Short-Term Memory (LSTM) and Bidirectional LSTM (BiLSTM), have demonstrated strong capabilities in capturing temporal dependencies in time-series data[13,14]. Beyond agricultural and spectroscopic applications, recent studies on data-driven monitoring of complex engineering systems have also demonstrated the importance of hybrid feature extraction and graph-based representation learning. For example, Fu et al. proposed a hybrid monitoring strategy combined with an improved graph convolutional network for fault detection and diagnosis of air handling units, showing that nonlinear feature extraction and graph-based learning can improve diagnostic performance under complex operating conditions[15]. Zhang et al. further integrated kernel Fisher discriminant analysis with an improved graph convolutional neural network to incorporate class-label information and spatial relationships for air-handling-unit fault diagnosis[16]. To further enhance model adaptability under complex conditions, researchers have incorporated attention mechanisms to dynamically assign importance to different time steps or feature dimensions, effectively suppressing redundant information caused by environmental disturbances and improving prediction accuracy and robustness[17–19]. For example, Song et al. proposed a quantitative regularization strategy for robust Vision Transformer-based remote sensing image classification, demonstrating that regularized attention-based representation can improve model stability when dealing with complex high-dimensional visual data[20]. Similarly, Causal-YOLO introduced causal feature alignment to distinguish invariant target-related features from spurious environmental interference in remote sensing object detection[21]. Although these studies were conducted in remote sensing scenarios, their methodological insights are relevant to near-infrared spectroscopy-based process monitoring, where spectral responses are also high-dimensional, nonlinear, and susceptible to environmental disturbances. Previous studies have validated the potential of deep learning methods for moisture detection in tea, black tea withering, and agricultural product drying under dynamic conditions[1,3,22]. Recent reviews indicate that combining near-infrared spectroscopy with deep learning represents a promising direction for online quality monitoring in food processing, especially in non-stationary and disturbance-prone industrial applications[23,24]. Nevertheless, for the typical long-duration and environmentally coupled white tea withering process, spectral prediction studies integrating temporal modeling with attention mechanisms remain limited, underscoring the need for further systematic investigation.

From the perspective of automation and intelligent control, existing white tea withering systems still face several practical limitations. First, most production lines are still operated using rule-based temperature and ventilation settings, while the withering degree is judged mainly by worker experience; therefore, the control loop is not closed by quantitative moisture information. Second, the long withering duration causes dynamic coupling among leaf phenotype, ventilation, ambient humidity, and spectral response, which makes static calibration models unstable when they are transferred from laboratory conditions to production lines. Third, online control requires a prediction model that is not only accurate but also interpretable and computationally feasible for industrial edge computers. Accordingly, the proposed STA-BiGRU-XGBoost framework was designed to address these limitations by integrating feature screening, spatial attention, temporal dependency modeling, and robust ensemble regression into a two-stage prediction pipeline.

Based on the aforementioned challenges and recent research progress, this study focuses on the long-duration, environmentally coupled white tea withering process and aims to develop a near-infrared spectral deep learning prediction framework that integrates temporal modeling with attention mechanisms. The main contributions of this paper are as follows:

(1) A systematic analysis of the correlation between dynamic moisture changes during white tea withering and near-infrared spectral features.

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(2) The novel proposal of a STA-BiGRU-XGBoost model that integrates spatiotemporal dual attention mechanisms, bidirectional GRU, and XGBoost to simultaneously enhance long time-series modeling, focus on critical information, and improve prediction stability under the investigated industrial noise conditions.
(3) A validation of the proposed method's prediction accuracy, stability, and interpretability under the investigated production-line conditions, providing technical support for moisture monitoring and intelligent control of the white tea withering process within the studied operational scenario.

Materials and methods

Data acquisition device

The white tea withering system comprises several key components, including a feed inlet, greening mechanism, conveyor belt, dehumidifying heat pump, exhaust fan, discharge outlet, and an infrared spectrometer, as illustrated in Fig. 1. Fresh tea leaves are introduced through the bottom inlet and transported by the greening mechanism onto the withering conveyor belt. The dehumidifying heat pump supplies temperature- and humidity-controlled hot air to the withering chamber, facilitating moisture removal while suppressing green odor formation. After 24–48 h of withering, the leaves are transferred to the discharge outlet, completing the process. The infrared spectroscopy moisture detection system, utilizing a MERS Measurement MS-580 near-infrared moisture analyzer, collects real-time, non-destructive moisture data from the tea leaves. The system operates within the 940–1550 nm wavelength range and is installed directly above the conveyor belt.

Sample preparation and data collection

Data collection was conducted from March 18 to May 18, 2025, at a white tea processing company in Wuyishan City. The fresh tea leaves used in the experiments were sourced from Zhenghe County, Nanping City, China. Fresh white tea leaves are primarily harvested between March and May each year, and most tea factories concentrate their withering processes during this period. It should be noted that tea leaves are not withered continuously throughout the year, resulting in minimal seasonal variations in the withering datasets. During tea production, the installed infrared spectrometer collected sample data from leaves in the greening mechanism. The near-infrared spectral data collection method is illustrated in Fig. 2. To assess the influence of operational temperature and humidity at different times on near-infrared spectra and prediction accuracy, a sampling interval of 5 min was adopted. To ensure consistent withering quality, the tea factory regulated the production workshop environment (ambient temperature: 20–35 °C, humidity: 45–60% RH), maintaining stable conditions without significant fluctuations. Withering process data were collected by an industrial control computer (with a sampling frequency of 10 Hz for each sensor, processed using a 5-minute moving average filter). The computer saved the processed data at the same interval, enabling real-time monitoring and precise control of variables throughout the withering process and ensuring that the collected data accurately reflected actual conditions. Table 1 provides detailed descriptions of the collected process variables, each carefully selected based on its significant impact on final quality and processing efficiency. The dataset was divided into three subsets: a training set for learning model parameters, a validation set for hyperparameter tuning, and a test set comprising two batches for evaluating prediction model performance.

During data acquisition, the near-infrared spectrometer collected spectral information in the wavelength range of 940–1550 nm. These wavelength-resolved spectra were mainly used for spectral characterization, preprocessing comparison, and identification of moisture-sensitive absorption regions. For online modeling and real-time prediction, the industrial NIR system further generated a synchronized scalar absorbance feature from the selected moisture-sensitive spectral response. This scalar feature was recorded by the industrial control computer together with the process variables, including hot air velocity, air duct temperature, and environmental parameters. In this study, this online scalar spectral absorbance feature was denoted as X_9. The adoption of the scalar absorbance feature X_9 was determined by both the industrial online acquisition configuration and the

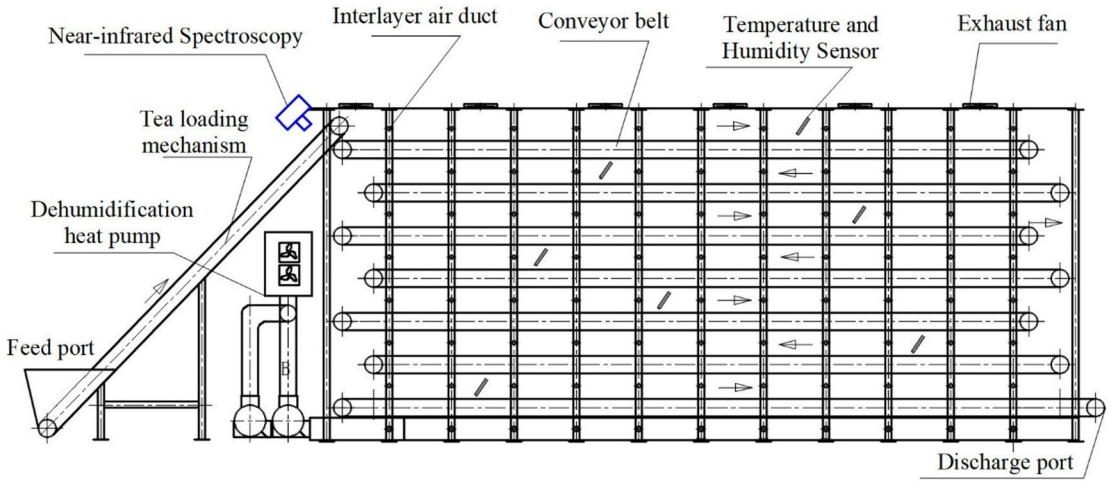
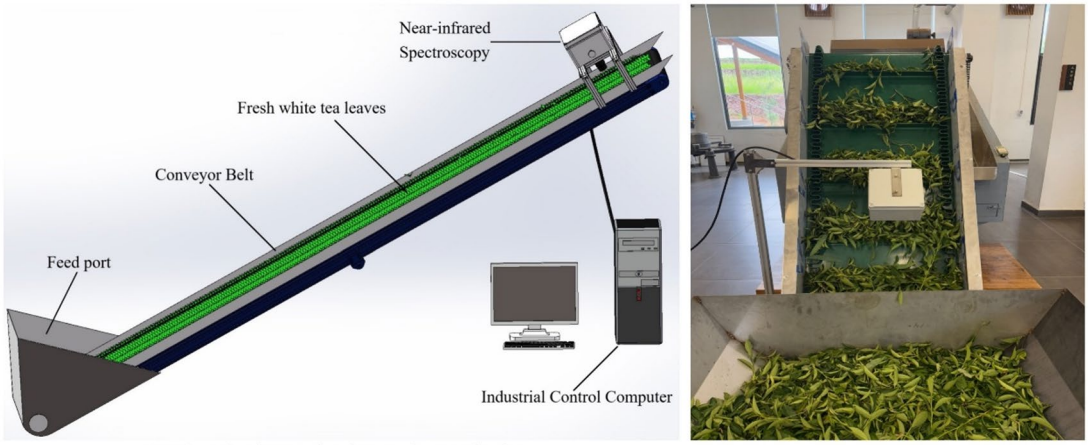


Fig. 1. Schematic of the white tea withering system.

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(a) Schematic of Near-Infrared Spectral Data Collection

(b) Schematic of Near-Infrared Spectral Working Environment Data Collection

Fig. 2. Schematic of near-infrared spectral data collection.

Serial No.	Variable names	Unit	Scope of the variables
X_1	Withering outdoor temperature	℃	10.0–40.0
X_2	Withering outdoor humidity	%	50.0–90.0
X_3	Withering indoor temperature	℃	20.0–40.0
X_4	Withering indoor humidity	%	50.0–80.0
X_5	Withering room heat pump outlet temperature	℃	20.0–50.0
X_6	Withering room hot air velocity	m/s	0–3.0
X_7	Withering room interlayer air duct outlet temperature	℃	20.0–40.0
X_8	Feed flow rate	kg/s	0–10.0
X_9	Near-infrared spectral absorbance	%	10–150
X_{10}	Discharge port temperature	℃	20.0–40.0

Table 1. Withering tea process variables.

real-time monitoring objective of this study. Although the full wavelength-resolved spectra over 940–1550 nm contain richer spectral information, direct ingestion of the complete spectrum would substantially increase the input dimensionality, model parameters, storage demand, and online inference burden. In contrast, the industrial NIR moisture analyzer used in the production line outputs a synchronized absorbance indicator derived from the moisture-sensitive spectral response. This scalar output is directly available to the industrial control computer and can be reliably aligned with process variables at the 5-min sampling interval. Therefore, X_9 was used as the spectral input feature for the proposed time-series prediction framework, while the full spectra were used mainly for spectral characterization, preprocessing comparison, and identification of moisture-related absorption regions. X_9 does not represent the complete high-dimensional spectrum from 940 to 1550 nm, but rather the scalar absorbance feature extracted from the NIR spectral response for real-time modeling.

Measurement of moisture content

The moisture content of fresh white tea leaves was determined using the direct drying method as specified by the Chinese National Standard GB5009.3-2016[25], the measurement of moisture content is shown in Fig. 3. Each sample was weighed before and after drying in an oven. The average moisture content from three measurements was used to calculate the moisture percentage with the following formula:

$$w = \frac{m - m_1}{m} \times 100\% \quad (1)$$

where m is the initial mass of the tea sample and m_1 is the mass after drying.

Dataset partition and data preprocessing

To improve reproducibility and avoid future-data leakage in time-series modeling, all samples were arranged according to their chronological acquisition order before dataset division. No random splitting was used. The dataset summary and chronological sample allocation are provided in Table 2. The high-dimensional spectral records and process variables were aligned using the same 5-min timestamp before model training.

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Fig. 3. The measurement of moisture content.

Subset	Chronological role	Sample size (*N*)	Input variable categories	Purpose
Training set	Earliest 70% of aligned records	12,499	Process variables + NIR-derived absorbance feature	Learning STA-BiGRU parameters
Validation set	Following 15% of aligned records	2,678	Same input features as training set	Hyperparameter tuning and early stopping
Test set 1	Next 7.5% of aligned records	1,339	Same input features as training set	Primary independent performance evaluation
Test set 2	Final 7.5% of aligned records	1,340	Same input features as training set	Secondary batch validation and robustness assessment
Total	March 18-May 18, 2025; 5-min interval	17,856	X1-X10 initially; X6, X7, and X9 after mRMR	Full production-line dataset

Table 2. Summary of dataset and chronological sample allocation.

Preprocessing algorithm	Mathematical definition	Description
MSC	$x_i(\lambda_j) = a_i + b_i \bar{x}(\lambda_j) + e_i(\lambda_j)$ $x_i^*(\lambda_j) = \frac{x_i(\lambda_j) - a_i}{b_i}$	a_i and b_i are the intercept and slope obtained by regressing the spectrum of the i-th sample against the mean reference spectrum. MSC corrects additive and multiplicative scattering effects.
SNV	$x_i^*(\lambda_j) = \frac{x_i(\lambda_j) - \mu_i}{s_i}$ $\mu_i = \frac{1}{p}\sum_{j=1}^{p} x_i(\lambda_j)\ s_i = \sqrt{\frac{1}{p-1}\sum_{j=1}^{p}[x_i(\lambda_j) - \mu_i]^2}$	μ_i and s_i are the mean and standard deviation of the i-th spectrum. SNV reduces baseline shift and scattering caused by particle size and optical path variation.
SG smoothing	$x_i^*(\lambda_j) = \sum_{k=-m}^{m} c_k x_i(\lambda_{j+k})$	$2m+1$ is the smoothing window length, and c_k represents the convolution coefficient obtained by local least-squares polynomial fitting. SG smoothing suppresses random spectral noise while preserving peak shape.

Table 3. Mathematical definitions of spectral preprocessing algorithms.

Preprocessing algorithm	Mathematical definition	Description
D1	$x_i^*(\lambda_j) = \frac{dx_i(\lambda_j)}{d\lambda}\|_{\lambda=\lambda_j}$	$x_i^*(\lambda_j) \approx \frac{x_i(\lambda_{j+1}) - x_i(\lambda_{j-1})}{\lambda_{j+1} - \lambda_{j-1}}$; if the wavelength interval is constant, then: $x_i^*(\lambda_j) \approx \frac{x_i(\lambda_{j+1}) - x_i(\lambda_{j-1})}{2\Delta\lambda}$

Due to external environmental factors and instrument interference, near-infrared spectroscopy systems are susceptible to noise. Therefore, appropriate spectral preprocessing techniques are crucial for mitigating these effects. In this study, four spectral preprocessing algorithms were applied: Multiplicative Scatter Correction (MSC), Standard Normal Variate (SNV), Savitzky–Golay smoothing (SG), and first-order derivative (D1). The four preprocessing algorithms were mathematically defined as follows. For the spectrum of the i-th sample, $x_i(\lambda_j)$ denotes the raw absorbance at the wavelength λ_j, where $j = 1,2,\ldots,p$, and p is the number of wavelength points. The mean reference spectrum is denoted as $\bar{x}(\lambda_j)$, and the preprocessed spectrum is

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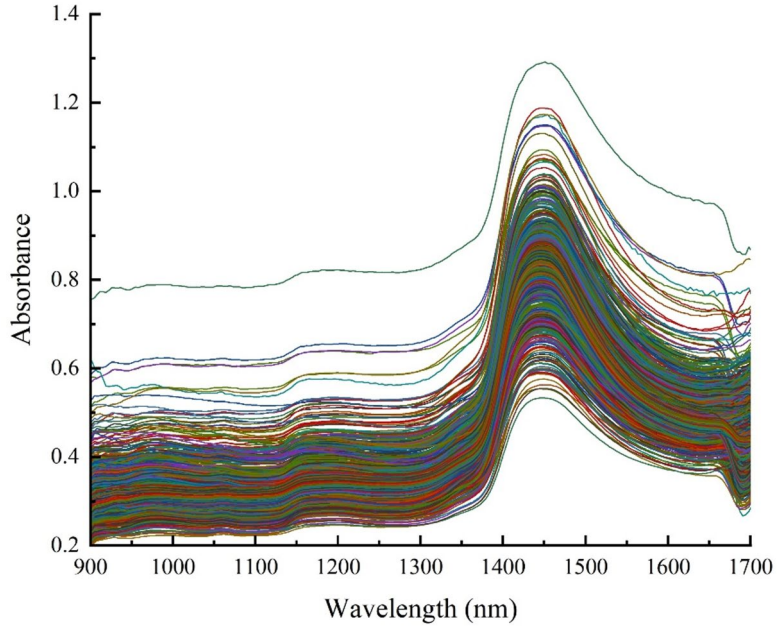


Fig. 4. Raw NIR spectral curves of white tea leaves.

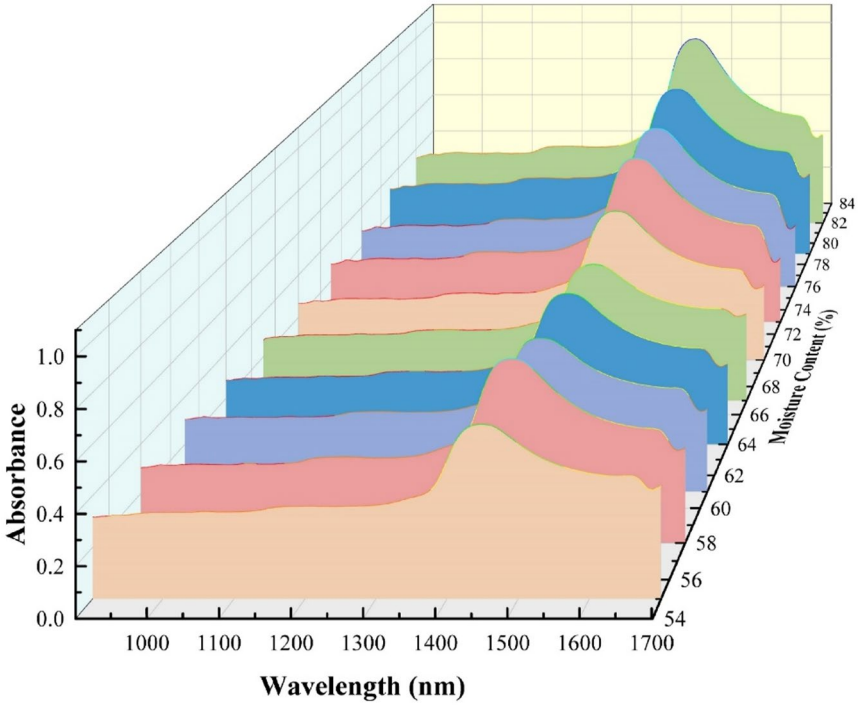


Fig. 5. NIR spectral curves after MSC preprocessing.

denoted as $x_i^*(\lambda_j)$. The mathematical definitions of the four spectral preprocessing algorithms are summarized in Table 3.

SNV and MSC correct for optical path variations, reducing scattering effects on spectral information. SG and D1 remove noise and reduce spectral errors caused by derivative-based background effects. These preprocessing methods help diminish external interference, improving prediction model accuracy and generalization. Figure 4 shows the raw near-infrared spectral curve of a fixed leaf sample. Significant absorption peaks were observed in the 1160–1220 nm and 1420–1480 nm bands, primarily associated with symmetric and asymmetric stretching vibrations of O–H bonds in water molecules[26,27]. After MSC preprocessing, absorption peaks in training samples

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with different moisture gradients became clearer with reduced scattering, enabling more accurate moisture content prediction, as shown in Fig. 5.

Prediction method determination

The white tea withering process occurs in a highly variable field environment where sensor sensitivity can be affected by external disturbances. To address this, a neural-network-based method was employed to process the infrared spectrometer data. Neural networks can be categorized into feedforward and recurrent types. Feedforward networks, lacking feedback and inhibition mechanisms, are unsuitable for this task. In contrast, recurrent networks, which incorporate feedback and inhibition, optimize the network through algorithms, making them more appropriate for this study.

Deep learning networks, especially those designed for time-series prediction, have demonstrated outstanding performance in modeling complex dynamic processes. In this study, the STA-BiGRU model was applied for the first time to predict moisture content during tea withering. The model integrates STA and BiGRU, enabling adaptive weighting of time steps and capturing long-term dependencies in moisture content sequences.

The withering process exhibits dynamic characteristics, including spatial and temporal dependencies, along with noisy and inconsistent data, which pose challenges for accurate prediction. The data characteristics can be summarized as follows:

- **Spatial dependence**: Heat and mass transfer phenomena lead to significant spatial correlations among variables.
- **Temporal dependence**: Moisture content is influenced by past values, and fluctuations in process parameters cause dynamic changes.
- **Data quality**: The complex environment results in noisy, missing, and inconsistent data, affecting model performance.

Given a set of input variables $X = \{x_1, x_2, \ldots, x_t\}$, where t represents time and x_t is the observed value at time t, the model predicts the key variables for future time periods. The prediction function can be expressed as:

$$[\hat{y}_{T+1}, \hat{y}_{T+2}, \cdots, \hat{y}_{T+\tau}] = \hat{f}(X, y_1, y_2, \cdots, y_T) \tag{2}$$

where $\hat{f}$ is a nonlinear prediction function, T represents the size of the historical data window, τ is the forecast horizon, y_T is the moisture content at time T, and $\hat{y}_{T+1}$ is the predicted moisture content at time $T+1$. In this study, the experimental design was a single-step prediction task, and therefore the forecast horizon was fixed as $\tau = 1$; multi-step rolling prediction is discussed as future work.

STA-BiGRU method

The STA-BiGRU method for moisture content prediction during white tea withering consists of four main components: feature selection using mRMR, a spatiotemporal attention mechanism, bidirectional GRU

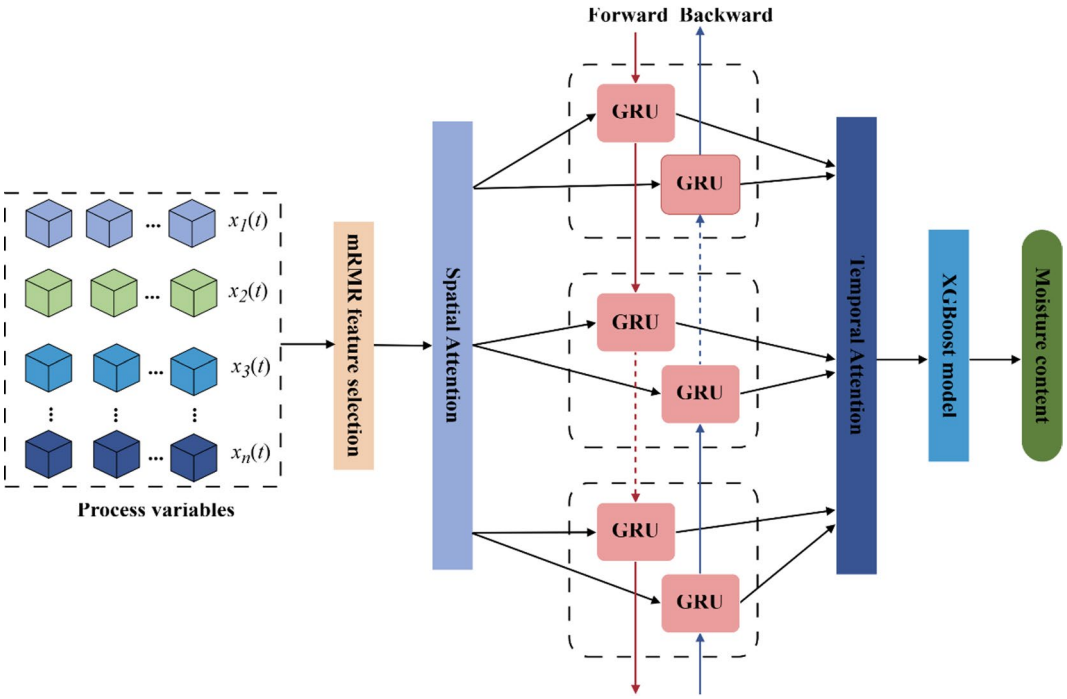


Fig. 6. Overall framework of the STA-BiGRU-XGBoost prediction model.

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modeling, and XGBoost-based moisture content prediction (MC). The overall framework is illustrated in Fig. 6. The key steps are as follows:

As shown in Fig. 6, the framework contains five sequential modules: (i) synchronized process and spectral data acquisition, in which environmental/process variables and NIR spectra are aligned by timestamp; (ii) preprocessing and mRMR feature selection, which suppress spectral noise and remove redundant variables; (iii) spatial attention, which assigns dynamic weights to input features at each time step; (iv) BiGRU with temporal attention, which extracts bidirectional temporal features and highlights key time steps; and (v) XGBoost regression, which maps the frozen attention-enhanced context vectors to final moisture content predictions.

During tea withering, high-dimensional variables collected by the processing system (e.g., environmental temperature, humidity, spreading thickness, and ventilation rate) increase model complexity and may reduce prediction accuracy. Previous studies have shown that the mRMR algorithm effectively removes redundant features while retaining key variables, improving model performance (with a 9% increase in R^3) and preserving the physical interpretability of selected features[28,29]. Therefore, the mRMR algorithm was adopted for feature selection in this study.

During the tea withering process, different variables exert varying levels of influence on moisture content at different time points. A spatial attention mechanism was employed to adaptively assign weights to each feature according to its correlation with moisture content, thereby enhancing the model's ability to focus on relevant variables. Let the input variables be denoted as $X = \{x_1(t), x_2(t), \ldots, x_n(t)\}$, where $t$represents time, $x_n(t)$denotes the observed value at time t, and n is the number of input features. For each time step t, the attention weight $a_i$of each feature is calculated as follows:

$$\begin{aligned} a_i &= \mathrm{softmax}(e_i) = \frac{\exp(e_i)}{\sum_{i=1}^{m} e_i} \\ e_i &= \mathrm{sigmoid}\left(W_s x_n + b_s\right) \end{aligned} \tag{3}$$

where $\mathrm{x_n}$ is the input vector of the n-th feature at time step t, W_s, and b_s are the parameters of the spatial attention layer, which compute the weighted sum of the input features as the context vector X':

$$\mathrm{X'} = \mathrm{A} \odot \mathrm{X} \tag{4}$$

where A is the attention weight vector and $\odot$ denotes the Hadamard product. The spatially weighted input features X' serve as the input to the bidirectional GRU at time step t, enabling the model to focus on features most relevant to the current moisture content prediction.

Specifically, the unnormalized score $\mathrm{e_i}$ is first generated by the spatial attention layer from the current feature vector and trainable parameters $\mathrm{W_s}$ and $\mathrm{b_s}$. The softmax operation then normalizes all scores across the feature dimension, ensuring that a_i is non-negative and that the sum of all feature weights at the same time step equals 1. Thus, Eq. (4) uses the learned attention vector A to rescale the original input features before they enter BiGRU.

GRU captures long-term dependencies in time-series data through update and reset gates, making it suitable for handling temporal correlations in tea withering. This study employs BiGRU, which combines forward and backward information, enhancing the model's understanding of sequential data. The key equations of GRU are as follows:

$$\begin{cases} z_t = \sigma(W_z[h_{t-1}, X_t^{'}] + b_z) \\ r_t = \sigma\left(W_r[h_{t-1}, X_t^{'}] + b_r\right) \\ \tilde{h}_t = \tanh(W_h[r_t \odot h_{t-1}, X_t^{'}] + b_h) \\ h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t \end{cases} \tag{5}$$

where z_t is the update gate controlling historical state retention, r_t is the reset gate adjusting historical information forgetting, h_{t-1} and h_t are previous and current hidden states, X_t' is the spatially weighted input, $\sigma(\cdot)$ is the sigmoid activation function, and $\odot$ is the Hadamard product.

Standard GRUs only consider forward information flow. This study uses BiGRU to capture long-term temporal dependencies in withering. The bidirectional structure allows the model to utilize both historical and future information, improving its ability to capture complex temporal dependencies. BiGRU consists of two GRU sub-networks running in opposite directions, capturing forward and backward dependencies, respectively. By concatenating forward and backward hidden states, BiGRU comprehensively represents moisture variation dynamics during withering:

$$\begin{cases} \overrightarrow{\mathrm{h}}_{\mathrm{t}} = \overrightarrow{\mathrm{GRU}}(\mathrm{h_{t\text{-}1}}, X_t') \\ \overleftarrow{\mathrm{h}}_{\mathrm{t}} = \overleftarrow{\mathrm{GRU}}(\mathrm{h_{t+1}}, X_t') \\ \mathrm{H_t} = \left[\ \overrightarrow{\mathrm{h}}, \overleftarrow{\mathrm{h}}\ \right] \end{cases} \tag{6}$$

where $\overrightarrow{h}_t$ and $\overleftarrow{h}_t$ are outputs of the forward and backward GRUs, respectively, and X_t' is the initial cell state (zero-initialized).

The tea withering process exhibits significant long-term dependencies and dynamic characteristics. As input sequence length increases, model performance tends to degrade. To address this, a spatiotemporal attention mechanism is employed to adaptively weigh the importance of different time steps. Let $H = [h_1, h_2, \ldots, h_n]$ represent hidden states from BiGRU; the temporal attention weight β_i for each time step is calculated as:

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$$\mathrm{l_i}=\mathrm{ReLU}(\mathrm{W_i h_i}+\mathrm{b_i}) \tag{7}$$

$$\beta_{\mathrm{i}} = \mathrm{softmax}\,(\mathrm{l_i}) = \frac{\mathrm{e}^{\mathrm{l_i}}}{\sum_{\mathrm{i}=1}^{\mathrm{n}} \mathrm{l_i}} \tag{8}$$

where W_i is a trainable weight matrix, b_i is the bias, and ReLU is the activation function. The context vector H' is computed as the weighted sum:

$$H' = \Sigma_{i=1}^{n} \beta_i h_i = \beta^T H \tag{9}$$

where $\beta = [\beta_1, \beta_2, \ldots, \beta_n]^T$ is the normalized temporal attention vector. The context vector H' is therefore the weighted sum of BiGRU hidden states.

mRMR feature selection for tea withering process variables

The tea withering system is influenced by numerous process variables; however, high-dimensional data can complicate network structure and reduce prediction accuracy. To address this, the mRMR algorithm was used to extract informative and highly correlated features from high-dimensional inputs. Given an input feature set $S = \{x_1(t), x_2(t), \ldots, x_n(t)\}$ and target variable $y(t)$, mRMR selects a feature subset that maximizes relevance to the target while minimizing redundancy. The objective function is:

$$mRMR = \max_S \left[\frac{1}{|S|} \sum_{X_i(t) \in S} I\left(X_i(t) : Y(t)\right) - \frac{1}{|S|^2} \sum_{X_i(t), X_j(t) \in S} I\left(X_i(t) : X_j(t)\right) \right] \tag{10}$$

where $I(\cdot)$ denotes mutual information. Features are incrementally added to maximize the objective until K features are selected, with K tuned via validation performance.

Applying mRMR to variables in Table 1, X_6 (hot air velocity), X_7 (air duct outlet temperature), and X_9 (spectral absorbance) showed the highest correlations and were selected as inputs for the deep learning model.

The use of the scalar NIR-derived feature X_9 represents an engineering trade-off between spectral information richness and industrial deployment feasibility. Full-spectrum input over 940–1550 nm may preserve more wavelength-resolved information and may allow a neural network to learn nonlinear interactions among adjacent wavelengths. However, direct use of the full preprocessed spectrum would also substantially increase the input dimensionality, model parameters, memory consumption, and inference latency. More importantly, high-dimensional full-spectrum modeling generally requires a larger and more diverse calibration dataset to avoid fitting spectral noise, baseline drift, and batch-specific variations.

For the GRU-based temporal modeling component, the computational cost at each time step is approximately proportional to:

$$O\left(3h(d+h)\right) \tag{11}$$

where d is the input dimension and h is the hidden dimension. In the present model, only three mRMR-selected variables, namely X_6, X_7, and X_9, were used as inputs. If the full preprocessed spectrum containing p wavelength points were directly used, the input dimension would increase from $d = 3$ to approximately $d = p + 2$, assuming that X_6 and X_7 are retained. This would increase the recurrent computation and memory demand considerably, which is not ideal for real-time deployment on an industrial edge computer.

Therefore, the scalar X_9 feature was adopted in the current production-line prediction framework to balance prediction accuracy, synchronization reliability, computational efficiency, and online deployment feasibility. Nevertheless, this strategy may not fully exploit all wavelength-level spectral interactions. Therefore, the comparison among full-spectrum input, selected multi-band spectral input, and scalar NIR-derived input will be further investigated in future work using larger cross-origin, multi-season, and multi-batch datasets.

Spatial attention method

Strong coupling exists among process variables during withering, and their importance to moisture content changes dynamically. A spatial attention method adaptively assigns weights to variables at each time step based on their correlation with moisture content. By dynamically focusing on relevant features, the model better captures spatial dependencies and interactions. The spatial attention mechanism is illustrated in Fig. 7.

For input variables $X = \{x_1(t), x_2(t), \ldots, x_n(t)\}$, the attention weight a_i for each feature at time t is calculated as:

$$a_i = softmax\,(e_i) = \frac{exp\,(e_i)}{\sum_{i=1}^{m} e_i} \tag{12}$$

$$e_i = sigmoid\,(W_s x_n + b_s) \tag{13}$$

where x_n is the input vector of the n-th feature, and W_s, b_s are spatial attention parameters. The context vector X' is:

$$X' = A \odot X \tag{14}$$

where $A = [a_1, a_2, \ldots, a_n]$. Spatially weighted features X' are fed into BiGRU at time step t.

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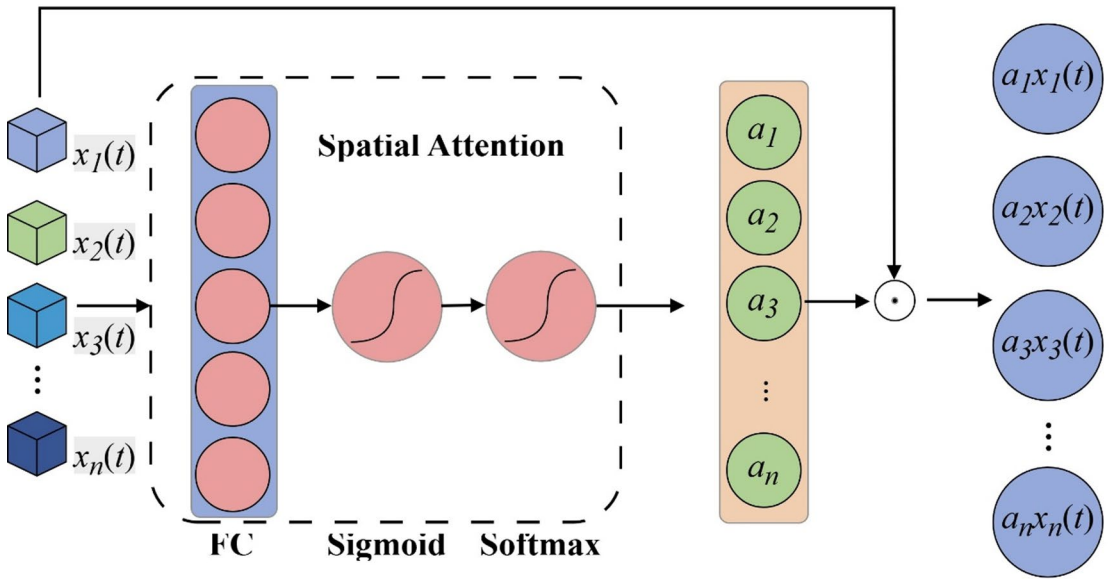


Fig. 7. Schematic diagram of the spatial attention method.

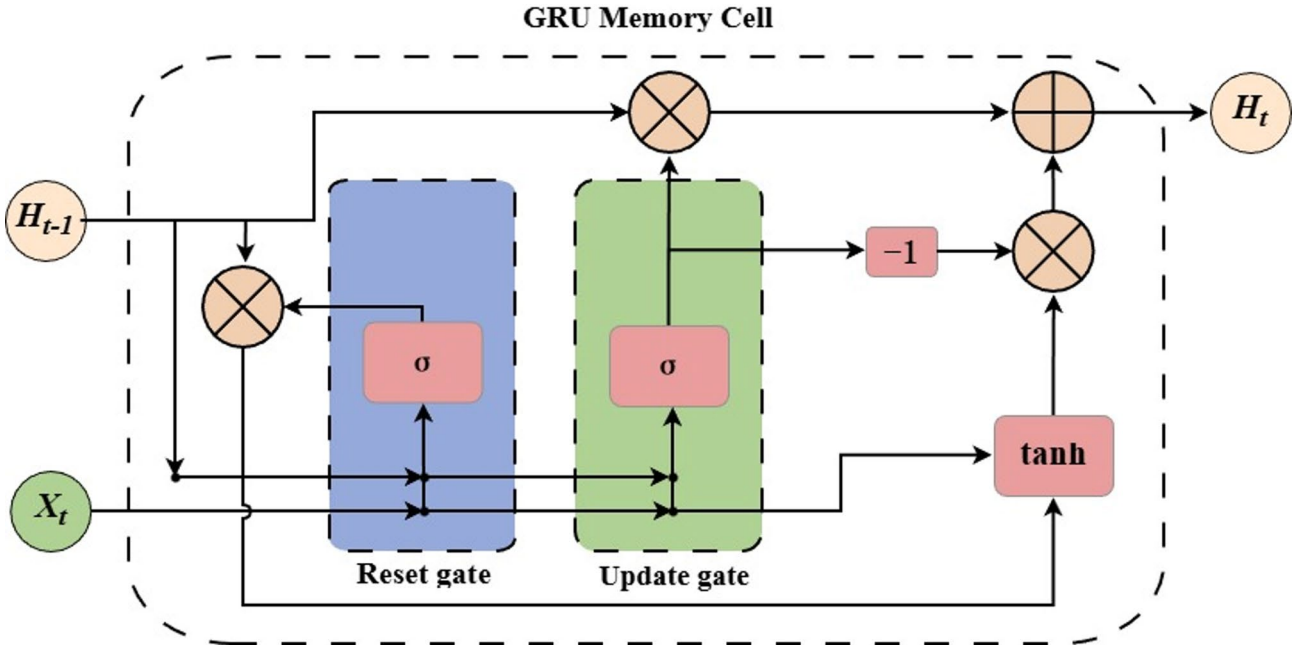


Fig. 8. Schematic diagram of the GRU.

GRU prediction model

GRU is a recurrent network composed of reset and update gates, efficiently controlling information storage and updating in time-series data. Its structure is shown in Fig. 8.

Traditional GRUs process information in one direction. This study uses BiGRU to capture long-term temporal dependencies by utilizing both past and future contexts. The BiGRU structure (Fig. 9) comprises two GRU layers running opposite directions.

The final BiGRU output is the concatenation of forward and backward hidden states, as in Eq. (6).

Time attention method

To handle long-term dependencies and input sequence length issues, a temporal attention mechanism adaptively weights different time steps based on their relevance to current prediction. The mechanism is illustrated in Fig. 10.

For hidden states $H = [h_1, h_2, \ldots, h_n]$ from BiGRU, temporal attention weight β_i is:

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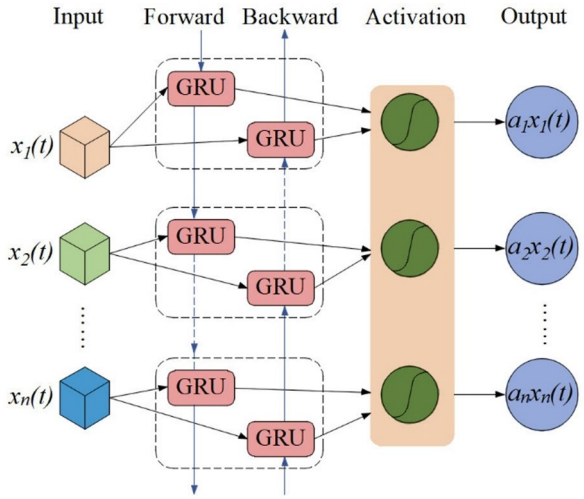


Fig. 9. The structure of BiGRU.

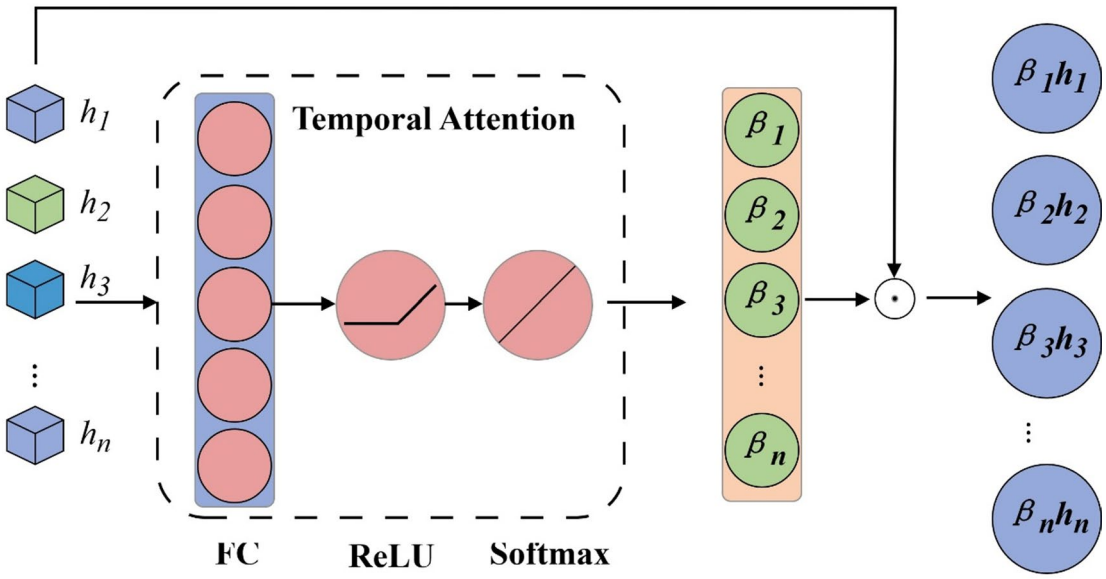


Fig. 10. The schematic diagram of the temporal attention method.

$$l_i = ReLU(W_i h_i + b_i) \tag{15}$$

$$\beta_i = softmax(l_i) = \frac{e^{l_i}}{\sum_{i=1}^{n} l_i} \tag{16}$$

where W_i is the training weight matrix, b_i is the bias term, and ReLU is the activation function.

The context vector H' is:

$$H' = \Sigma_{i=1}^{n} \beta_i h_i = \beta^T H \tag{17}$$

where $\beta = [\beta_1, \beta_2, \ldots, \beta_n]^T$ is the normalized temporal attention vector. This vector summarizes relevant temporal information for final prediction through a weighted summation of hidden states.

XGBoost algorithm

To enhance generalization and robustness against noise and harsh conditions, XGBoost is integrated as a post-processing step after STA-BiGRU. XGBoost is an efficient ensemble learning algorithm[30] that accelerates training via multithreading and distributed computing. It optimizes model parameters by minimizing a loss function comprising training error and regularization:

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$$Obj\,(\phi\,) = \sum\nolimits_{i=1}^{n} l(y_i, \widehat{y}_i) + \sum\nolimits_{k=1}^{K} \Omega\ (f_k) \tag{18}$$

where ϕ denotes model parameters, l is the loss function, and $\Omega\ (f_k)$ regularizes the k-th tree:

$$\Omega\ (f_k) = \gamma\, T + \frac{1}{2}\lambda \sum\nolimits_{j=1}^{T} \omega_j^{\,2} \tag{19}$$

with T leaf nodes, leaf weights ω_j, leaf penalty γ, and L2 regularization λ. Minimizing this objective reduces overfitting and improves generalization.

The proposed STA-BiGRU-XGBoost model was trained as a two-stage pipeline rather than an end-to-end jointly optimized network. In Stage 1, the STA-BiGRU backbone was trained using the training set, while the validation set was used for early stopping and hyperparameter selection. After convergence, the backbone parameters were frozen and the attention-enhanced context vector H' was extracted for each sample. In Stage 2, these frozen context vectors were used as inputs to train XGBoost independently, and the XGBoost hyperparameters were optimized on the validation set. This data flow was adopted to improve reproducibility and to reduce the risk of overfitting caused by joint optimization on a limited industrial dataset.

The two-stage design was adopted intentionally rather than as a simplified substitute for end-to-end optimization. In principle, end-to-end joint optimization may allow the feature extractor and the regression head to be updated simultaneously. However, for the present industrial dataset, such a strategy would introduce a larger number of trainable parameters and may increase sensitivity to batch-specific spectral drift, sensor noise, and short-term process disturbances. This issue is particularly relevant for long-duration white tea withering, where spectral responses and process variables are affected by environmental fluctuations and fresh-leaf heterogeneity.

Therefore, temporal representation learning and nonlinear regression were separated in the proposed framework. The STA-BiGRU backbone was first used to learn smoothed spatiotemporal representations under validation-based early stopping. After convergence, the backbone parameters were frozen, and the extracted context vectors were used as fixed high-level descriptors for XGBoost regression. The XGBoost stage further introduced shrinkage, row sampling, column sampling, and L1/L2 regularization, which helped reduce model sensitivity to noisy fluctuations and outliers in industrial process data. This design was intended to improve reproducibility and reduce the risk of overfitting under limited production-line samples.

XGBoost is applied to the output of the attention-enhanced BiGRU (Fig. 11) to handle noise and outliers common in withering environments. The final prediction is:

$$\widehat{y}_i = \sum\nolimits_{k=1}^{K} f_k\left(\boldsymbol{H}'\right) \tag{20}$$

where H' represents the context vector from the attention mechanism, and K represents the number of trees.

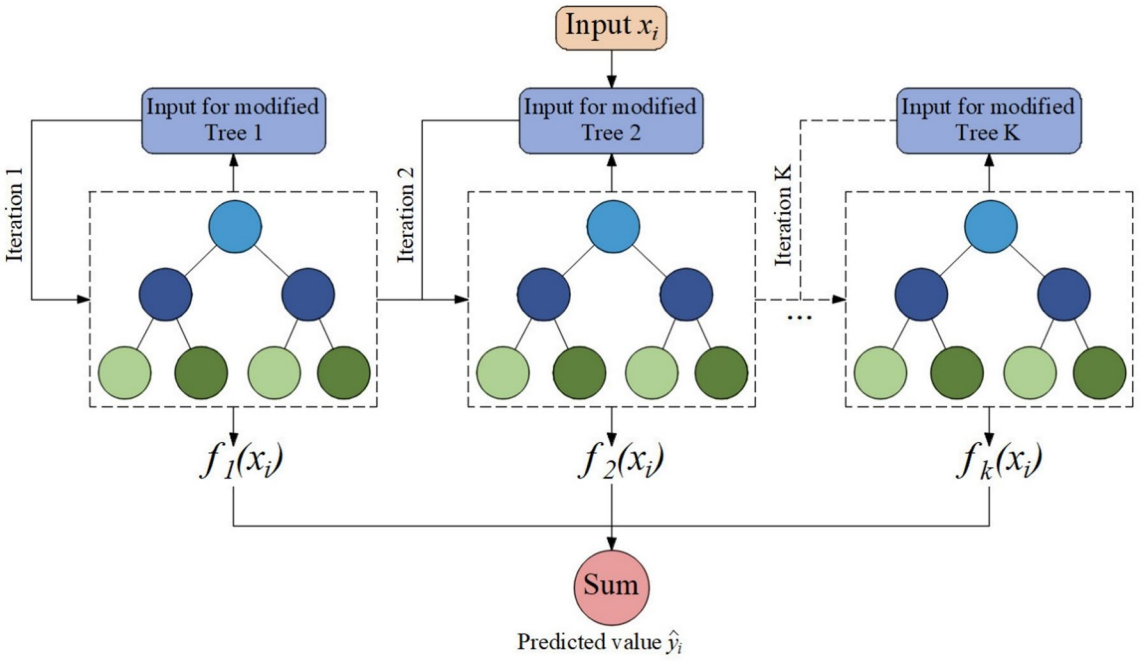


Fig. 11. Diagram of the XGBoost algorithm.

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Model evaluation metrics

Three metrics evaluate model performance: Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and Coefficient of Determination (R^2):

$$RMSE = \sqrt{\left(\sum_{j=1}^{n} (y_j - \widehat{y}_j)^2 / n\right)} \tag{21}$$

$$MAE = \sum_{j=1}^{n} |\widehat{y}_j - y_j| / n \tag{22}$$

$$R^2 = 1 - \frac{\sum_{j=1}^{n} (\widehat{y}_j - y_j)^2}{\sum_{j=1}^{n} \left(y_j - \bar{y}_j\right)^2} \tag{23}$$

where n is sample count, y_j and $\widehat{y}_j$ are true and predicted values, and $\bar{y}$ is the true mean.

Results and discussion

Experimental setup

The proposed STA-BiGRU-XGBoost model was compared with several deep learning models:

- LSTM: Advanced RNN handling long-term dependencies.
- GRU: Simplified LSTM variant with fewer parameters.
- BiGRU: Bidirectional GRU capturing past and future context.
- TA-BiGRU: BiGRU with temporal attention.
- STA-BiGRU: BiGRU with spatiotemporal attention.
- STA-BiGRU-XGBoost: STA-BiGRU integrated with XGBoost for noise robustness.

Hyperparameters were tuned via validation performance; configurations are listed in Table 4. For a fair comparison, all baseline models used the same chronological dataset split and the same three mRMR-selected input features (X6, X7, and X9). No baseline model was trained with the full original variable set. Therefore, performance differences among LSTM, GRU, BiGRU, TA-BiGRU, STA-BiGRU, and STA-BiGRU-XGBoost can be attributed mainly to the modeling architecture and the fusion strategy rather than to inconsistent input information.

The experiments were conducted on an industrial computer equipped with an Intel i7-1165G7 CPU (2.80 GHz) and 32.0 GB of RAM. The improved STA-BiGRU-XGBoost method and other algorithms were implemented using TensorFlow 2.19 and Python 3.13.0. The XGBoost model was built using the XGBoost 3.0 library.

To evaluate edge-deployment feasibility, the computational burden was analyzed separately for training and inference. Training is performed offline, whereas online monitoring only requires forward inference. For a

Prediction methods	Hidden Units	Dropout	Lr	Batch size	Epochs	Training strategy
Part A: deep learning backbone models.						
LSTM	256	0.35	0.001	128	100	Early Stopping (Patience = 15)
GRU	256	0.35	0.001	128	100	Early Stopping (Patience = 15)
BiGRU	256	0.35	0.001	128	120	Early Stopping (Patience = 20)
TA-BiGRU	256	0.35	0.001	128	120	Early Stopping (Patience = 20)
STA-BiGRU	256	0.35	0.001	128	120	Early Stopping (Patience = 25)
STA-BiGRU-XGBoost	256	0.35	0.001	128	120	Early Stopping (Patience = 25)
Hyperparameter	Baseline		Search Space		Purpose	
Part B: XGBoost hyperparameter tuning (for STA-BiGRU-XGBoost ensemble)						
n_estimators	300		[100,200,300,400,500]		Number of boosting rounds	
max_depth	6		[3,6,9]		Tree complexity and interaction depth	
learning_rate (η)	0.05		[0.05,0.1,0.2]		Shrinks tree contributions	
subsample	0.8		[0.7,0.8,0.9,1.0]		Row sampling per tree	
colsample_bytree	0.8		[0.7,0.8,0.9,1.0]		Feature sampling per tree	
reg_alpha (L1)	0		[0,1e-5,1e-2]		L1 regularization on leaf weights	
reg_lambda (L2)	1		[1,1.5,2]		L2 regularization on leaf weights	

Table 4. Hyperparameter configurations for all comparative models. Early stopping: Applied to all deep learning models based on validation loss plateauing. Patience (epochs) is set per model complexity. Optimizer: Adam optimizer is recommended for all deep learning models. XGBoost tuning: The ensemble model's XGBoost component undergoes independent hyperparameter optimization using the defined search space (e.g., via RandomizedSearchCV) to minimize validation RMSE.

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Methods	RMSE	MAE	R^2
LSTM	0.6962	0.5839	0.5162
GRU	0.6558	0.5072	0.6258
BiGRU	0.4824	0.3813	0.6731
TA-BiGRU	0.2465	0.2045	0.8141
STA-BiGRU	0.1325	0.1099	0.9533
STA-BiGRU-XGBoost	0.0920	0.0772	0.9806

Table 5. Experimental results on the tea withering dataset.

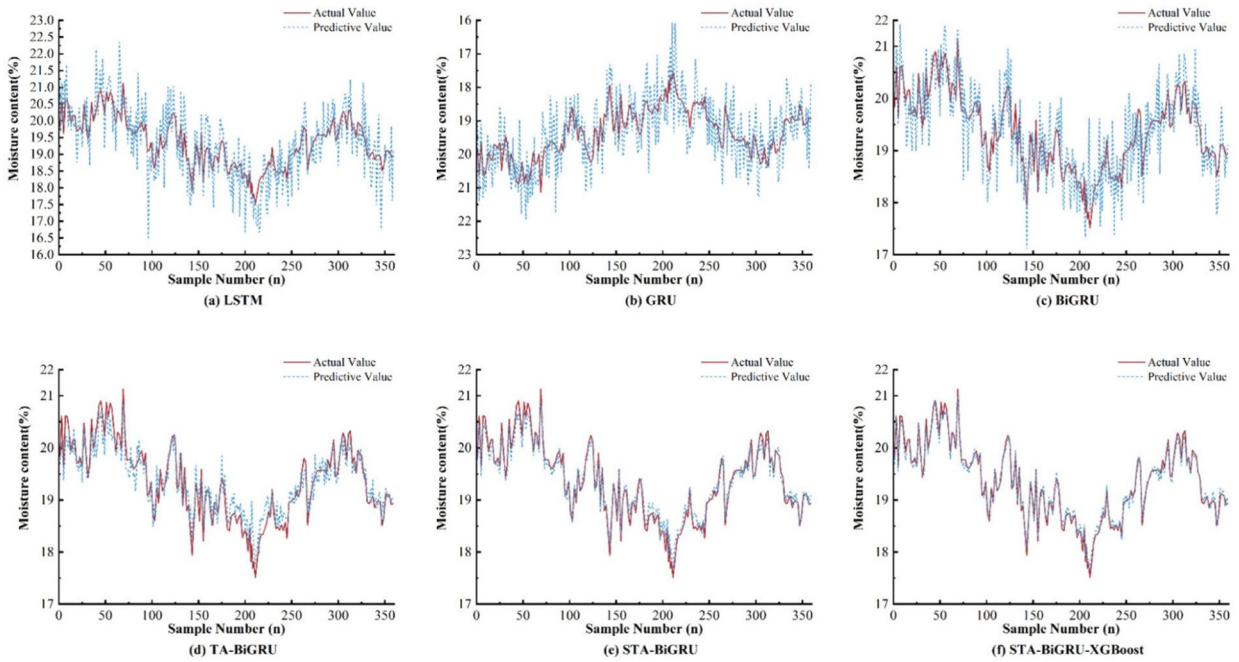


Fig. 12. Comparison of different moisture content prediction models.

historical window length L, input dimension d, hidden dimension h, and K XGBoost trees with maximum depth D, the approximate inference complexity is $O(L(dh + h^2) + KD)$. Because the sampling interval is 5 min and only three selected inputs are used after mRMR, the online inference burden is substantially lower than the sampling period, making the framework suitable for deployment on industrial edge computers after model training and parameter freezing.

Experimental results

Prediction performance comparison

Table 5 presents experimental results on the real-world withering dataset. STA-BiGRU-XGBoost achieved the best performance with RMSE = 0.0920, MAE = 0.0772, and R^2 = 0.9806, significantly outperforming all baseline models. LSTM and GRU showed higher errors and lower R^2, while BiGRU improved performance by capturing bidirectional context. TA-BiGRU further reduced error via temporal attention. STA-BiGRU, with spatiotemporal attention, achieved lower errors (RMSE = 0.1325, MAE = 0.1099) and higher R^2 (0.9533). The integration of XGBoost enhanced robustness, yielding the lowest errors and highest R^2.

The contribution of the XGBoost stage can be further observed by comparing STA-BiGRU with STA-BiGRU-XGBoost under the same chronological data split and the same mRMR-selected input features. Compared with STA-BiGRU, STA-BiGRU-XGBoost reduced RMSE from 0.1325 to 0.0920 and MAE from 0.1099 to 0.0772 on Test set 1. These reductions correspond to approximately 30.6% in RMSE and 29.8% in MAE. Meanwhile, R^2 increased from 0.9533 to 0.9806. Since both models used the same input variables and the same training/validation/testing protocol, the improvement can be mainly attributed to the XGBoost regression stage and its regularized ensemble learning capability.

Prediction and error curve analysis

Figure 12 compares actual vs. predicted moisture content curves for all models. LSTM and GRU exhibit considerable deviations and poor trend tracking, consistent with large error fluctuations in Fig. 13. BiGRU better captures temporal dynamics via bidirectional memory. TA-BiGRU reduces error further by focusing on

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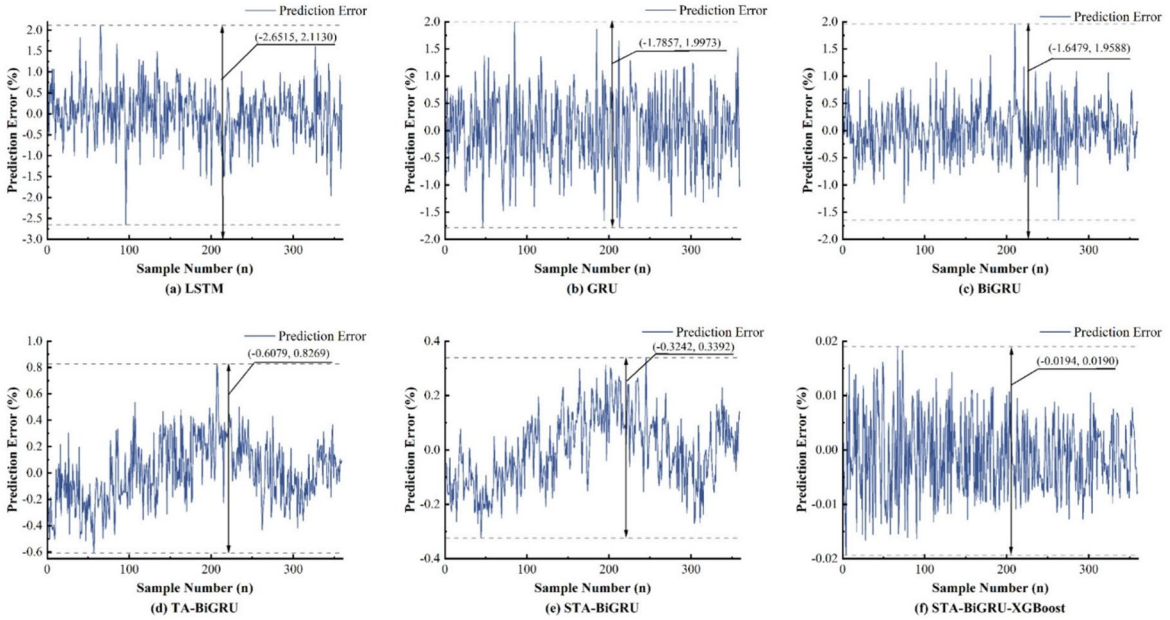


Fig. 13. Comparison of the errors of different moisture content prediction models.

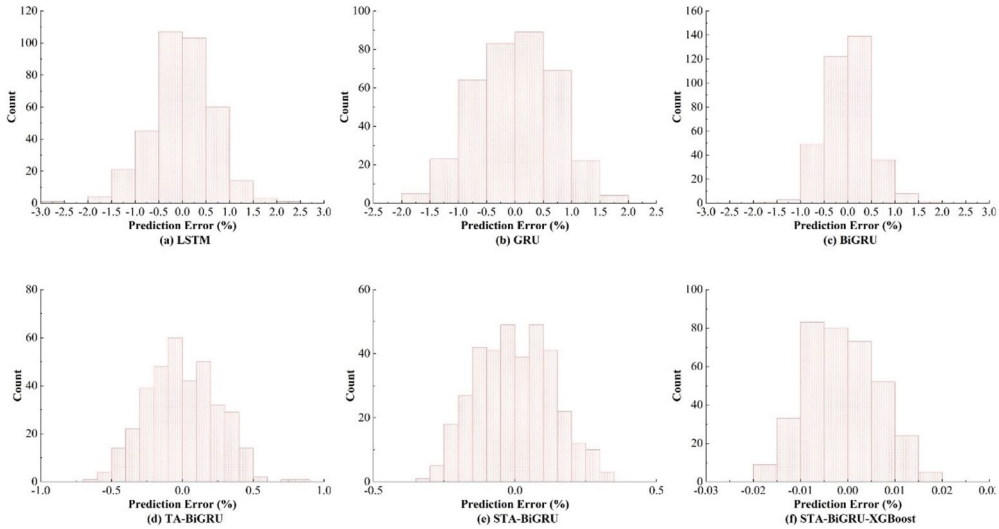


Fig. 14. Error probability distribution curves of different moisture content prediction models.

relevant time steps. STA-BiGRU, with spatial attention, improves performance, reducing prediction error by 11.2% relative to TA-BiGRU. STA-BiGRU-XGBoost shows the closest agreement with actual curves, with an average error reduction of 55.5% compared to STA-BiGRU, meeting on-site assessment requirements.

Robustness and reliability assessment

Figure 14 displays prediction error probability distributions for all models. STA-BiGRU-XGBoost exhibits the most concentrated distribution centered near zero, indicating superior reliability. Figure 15 shows error boxplots; STA-BiGRU-XGBoost has the narrowest interquartile range and fewest outliers, indicating improved error stability under the investigated production-line conditions.

Validation on a secondary test set

A secondary independent test set validated model reliability (Table 6). Results align with initial findings: attention-based models and the hybrid ensemble outperform conventional ones. STA-BiGRU-XGBoost again achieves the highest accuracy (RMSE = 0.0980, MAE = 0.0643, R^2 = 0.9879), supporting its effectiveness on the two temporally separated test sets used in this study.

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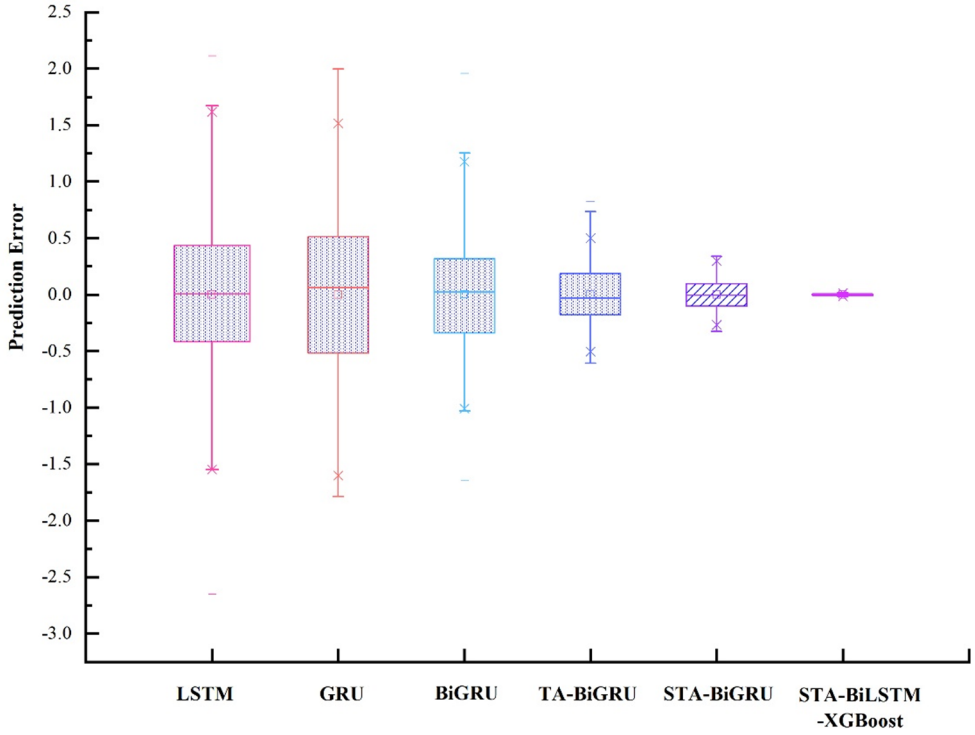


Fig. 15. Boxplots of predict error for different models.

Methods	RMSE	MAE	R^2
LSTM	0.6610	0.5721	0.5058
GRU	0.6414	0.5145	0.6147
BiGRU	0.4712	0.3950	0.6542
TA-BiGRU	0.2314	0.2158	0.8244
STA-BiGRU	0.1348	0.1102	0.9590
STA-BiGRU-XGBoost	0.0980	0.0643	0.9879

Table 6. Results on secondary tea withering dataset.

A similar improvement was observed on the secondary independent test set. Compared with STA-BiGRU, STA-BiGRU-XGBoost reduced RMSE from 0.1348 to 0.0980 and MAE from 0.1102 to 0.0643, corresponding to reductions of approximately 27.3% and 41.7%, respectively. R^2 also increased from 0.9590 to 0.9879. The consistent improvement across two temporally separated test sets indicates that the two-stage hybrid strategy improved generalization performance under the investigated production-line conditions, rather than merely fitting one specific test subset.

Industrial robustness to nonlinear drift

Although the proposed STA-BiGRU-XGBoost model achieved satisfactory prediction performance under the production conditions investigated in this study, the industrial robustness of near-infrared-based moisture prediction should be considered carefully. In practical white tea production, fresh leaves collected from different geographic origins, cultivars, harvesting batches, and weather conditions may exhibit differences in leaf thickness, maturity, biochemical composition, surface moisture, and internal water distribution. These phenotypic variations can induce nonlinear spectral drift, even when the measured moisture contents are similar. Such drift may affect the stability and transferability of a prediction model if it is trained only on data from a limited production region or season.

In the present study, several strategies were adopted to improve model robustness under dynamic production conditions. First, mRMR-based feature screening was used to remove redundant and weakly relevant variables, thereby reducing the risk of introducing unstable inputs. Second, the spatial attention mechanism adaptively assigned weights to different input features, allowing the model to emphasize moisture-related variables while suppressing less informative disturbances. Third, BiGRU and temporal attention were used to capture temporal dependencies during the long-duration withering process, which helped the model account for dynamic changes

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in spectral and environmental responses. Finally, XGBoost was introduced as the final prediction module to improve robustness through tree-based ensemble learning and regularization.

Nevertheless, it should be noted that the dataset used in this study mainly covered the production season, processing conditions, and fresh-leaf source region investigated in the experiment. Therefore, although the chronological validation and independent test sets can demonstrate the model's generalization ability within the studied production scenario, they cannot fully prove universal robustness across different geographic origins, cultivars, and extreme weather conditions. For broader industrial deployment, additional cross-origin, multi-season, and multi-weather datasets should be collected to further evaluate and improve model transferability.

Future work should focus on constructing a larger database containing fresh leaves from different tea-producing regions, cultivars, harvest periods, and weather conditions. In addition, practical model-maintenance strategies, such as periodic recalibration, calibration transfer, domain adaptation, and transfer learning, should be incorporated to compensate for nonlinear spectral drift caused by fresh-leaf phenotype variation and environmental changes. These strategies are expected to further enhance the robustness and applicability of near-infrared spectroscopy combined with deep learning for real-time moisture monitoring in industrial white tea withering.

Interpretability of experiments

To enhance interpretability, attention weight matrices were visualized as heatmaps (Figs. 16 and 17). Spatial attention (Fig. 16) shows weights across process variables over time. Variables X_6 (hot air velocity), X7 (air duct outlet temperature), and X_9 (spectral absorbance) exhibit consistently high weights (darker hues), indicating critical roles in withering. Variable X8 (feed flow rate) shows lower weights (lighter hues), aligning with known process dynamics. In industrial settings, increased X_6 and X_7 accelerate moisture evaporation, affecting spectral absorption—validating the model's focus on physically meaningful variables.

Temporal attention (Fig. 17) illustrates weight distribution across time steps. The current step (t) receives the highest weight (0.46), emphasizing its importance for instantaneous prediction. Weights decay for earlier steps, demonstrating adaptive focus on recent relevant information while retaining past context.

The spatiotemporal attention mechanisms not only improve performance but also provide insight into model decisions. Spatial heatmaps validate alignment with domain knowledge, while temporal distributions reveal how historical data is weighted, enhancing model transparency and credibility.

Conclusion

This study developed and validated an improved STA-BiGRU-XGBoost deep learning model for real-time moisture content prediction under the investigated white tea withering conditions. The model addresses key challenges in long-duration industrial monitoring via a multi-layered architecture: mRMR feature selection ensures interpretability; spatiotemporal attention adaptively focuses on relevant variables and time steps; BiGRU captures long-term dynamic dependencies; and XGBoost enhances generalization in noisy environments. Experimental results demonstrate that STA-BiGRU-XGBoost outperforms baseline models (LSTM, GRU, BiGRU, and attention variants) in accuracy (RMSE, MAE), goodness-of-fit (R^2), and error stability. Visualization

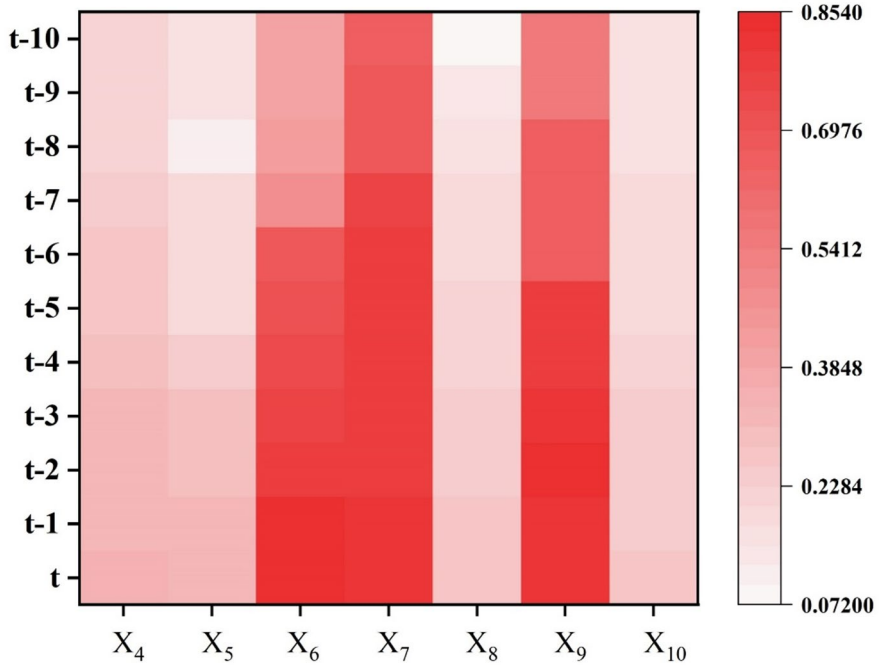


Fig. 16. Hot map of the spatial attention weight.

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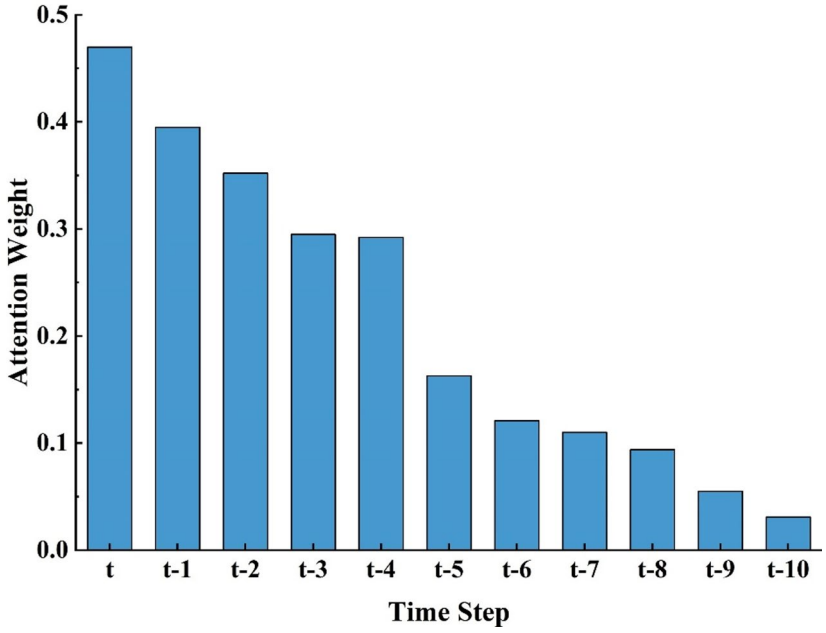


Fig. 17. The temporal attention weight.

of spatial attention weights confirms that model decisions align with core physical mechanisms (e.g., hot air velocity and temperature driving moisture evaporation), improving the transparency and interpretability of the prediction process.

It should also be acknowledged that the robustness of the proposed model was validated mainly under the production conditions and fresh-leaf sources considered in this study. In actual industrial applications, differences in geographic origin, cultivar, harvesting season, and weather conditions may cause nonlinear spectral drift and phenotypic variation in fresh leaves, which could affect model transferability. Therefore, future studies will further expand the dataset by incorporating cross-origin, multi-season, and multi-weather samples. Periodic recalibration, calibration transfer, domain adaptation, and transfer learning will also be explored to improve the long-term robustness and generalization ability of the proposed model in practical white tea production.

However, the current model is optimized for single-step prediction; its multi-step rolling prediction capability remains unexplored. Future work should: (1) design multi-step prediction architectures and explore decomposition methods for moisture content time series; (2) integrate multimodal data (e.g., visible light, hyperspectral imaging) to comprehensively characterize withering quality transformations; and (3) extend the framework to other agricultural processes with similar long-duration, strongly coupled characteristics to verify generalizability.

Data availability

The near-infrared spectral data and the code used for deep learning modeling in this study are available from the corresponding author upon reasonable request. The data are not publicly available due to confidentiality agreements with the tea manufacturer.

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Author contributions

Conceptualization, W.T. and B.C.; methodology, W.T. and B.G.; software, X.Y. and L.C.; validation, Y.H., R.L., J.R., and W.T.; investigation, L.C. and J.R.; data curation, Y.H. and B.G.; writing—original draft preparation, W.T. and B.C. All authors have read and agreed to the published version of the manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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